Fiscal Year:	FY 2021	**Task Last Updated:**	FY 01/13/2021
PI Name:	Christenson, Lane Ph.D.		
Project Title:	Female Reproductive Health: Space Flight Induced Ovarian and Estrogen Signaling Dysfunction, Adaptation, and Recovery		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	01/25/2018	**End Date:**	01/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 1/31/2022 per NSSC information (Ed., 1/22/2021) NOTE: End date changed to 1/31/2021 per F. Hernandez/ARC (Ed., 2/3/2020) NOTE: End date changed to 1/31/2020 per NSSC information (Ed., 3/12/19)		
Key Personnel Changes/Previous PI:	December 2020 report: No Key Personnel Changes.		
COI Name (Institution):	Ronca, April Ph.D. (NASA Ames Research Center) Alwood, Josh Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	NNX15AB48G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Continuation of "Female Reproductive Health: Space Flight Induced Ovarian and Estrogen Signaling Dysfunction, Adaptation, and Recovery" with the grant number NNX15AB48G and Principal Investigator (PI) Lane K. Christenson, Ph.D. (Previous PI was Joseph Tash, Ph.D.) In view of NASA's plans to conduct multigenerational studies on rodents in long-term space flight, these observations raise important fundamental questions on maintenance of female reproductive health that need to be addressed prior to conducting complex multigenerational studies in mammals under space flight conditions. As a result of our previous findings, specific important knowledge gaps now exist regarding the impact of space flight on female reproductive health. The central hypothesis that we will test is that space flight elicits direct and/or indirect effects at the ovarian level to cause infertility in adult female mice. This grant application combines the expertise of Dr. Christenson in ovarian physiology, and Dr. Ronca in pregnancy and early development, to address this central hypothesis. Additionally, these studies will address an important secondary hypothesis that the loss in uterine estrogen receptor expression will also occur in bone and this may exacerbate the observed bone mass loss during space flight (Dr. Alwood as collaborator). To address these hypotheses the following specific aims are proposed: Aim 1. Determine whether short-term (~10 d) space flight-disrupted ovarian function is due to a direct effect at the ovarian level. The working hypothesis, based on our preliminary data, is that disrupted ovarian function is due to a failure of the periovulatory follicle to ovulate and form a functional corpus luteum. To test this hypothesis, mice will undergo a timed follicular stimulation protocol followed by evaluation of established temporally expressed ovarian markers, ovulation, and luteal development. This study will for the first time determine if space induced loss of ovarian function is due to an intrinsic ovarian defect or is a problem with hypothalamic/pituitary support. Aim 2. Determine if adult female mice recover ovarian function during extended (~60 d) flight time. Currently, it is unknown whether females will recover fertility following long-term (i.e., ~60 d) acclimation to space flight, or will they remain infertile or exhibit more pronounced defects in ovarian/uterine function. To address this question, mice will undergo a similar follicular stimulation and ovulatory protocol in flight but after ~60 d, as described in Aim 1, and fertility and tissue biomarkers will be evaluated. Aim 3. Determine if adult female mice exposed to space flight are fertile upon return to Earth. The working hypothesis for this aim is that upon return to Earth, fertility will return. To test this hypothesis, flight mice will be exposed to male mice during a recovery period upon return to Earth, and regain of mating reproductive behavior, estrous cyclicity, early embryo implantation, and health and development of F1 offspring will be evaluated. These studies will provide the first ever examination of long-term space exposure on the female reproductive behavior/physiology, and germ-line stability. In all 3 specific aims, collection of uterine and bone (femur) tissues will allow us to address our secondary hypothesis by determining long-term space flight effects on estrogen receptor expression. Ultimately, findings of the proposed studies will provide critical foundational data to support multigenerational mammalian studies in space, and importantly, aid in our understanding of the impact space flight has on female astronaut health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Fertility is a general measure of fitness of a species on Earth. Disruptions to fertility have both short and long term effects on the health on the individual or the ability of a species to maintain critical numbers, respectively. Earth species have adapted to changes in environment over millennia and it is these adaptations that allow species to survive and flourish on Earth. The reproductive system of both female and male mammals are well tuned to sense environmental changes. These environmental responses can then either increase or decrease fertility appropriately. The effects of weightlessness on female ovarian function we currently are studying are critical as we consider longer term flights and the overall health of these individuals upon return to Earth. The fact that ovarian estrogen plays a key metabolic role in almost all tissues of the body also highlights the importance of understanding how female fertility is influenced by exposure to space. The Earth benefits of this research may yield new yet undiscovered mechanisms controlling ovarian function, steroidogenesis, and female fertility. Our research has led us to adapt an approach to monitor estrous cyclicity in rodents; we believe this methodology has the potential to impact all female based NASA flight research projects due to our ability to accurately determine the stage of the estrous cycle and thus predict ovarian estrogen levels.
Task Progress:	The experimental plan has been refined and has taken into consideration modifications that were necessary due to the Dragon Space Capsule berth time and crew time constraints. During this last year (2020) we submitted one abstract to American Society for Gravitational and Space Research (ASGSR) 2020 virtual meeting and an updated version to a regional K-INBRE (Kansas Idea Network of Biomedical Research Excellence) meeting occurring January 14-15, 2021 on a virtual platform, and a have a revised paper in review. In total we have 160 female mice of which 40 were exposed to microgravity on the International Space Station (ISS) for >30 days from these two studies, CASIS – Rodent Research Reference Mission 1 as well as the recent RRRM2 mission. These animals are providing key preliminary data, with respect to cyclicity as well as to how husbandry of the animals (i.e., the presence of the hut or lack of the hut) impacts the female endocrine system. Currently, as our flight is delayed until the "Hut versus No Hut" experiment is completed next year, we are excited that our analyses here will provide valuable insight into this important question. We currently have a revised paper in review that describes the effects of space flight on estrous cyclicity, and ovarian steroidogenic gene expression.
Bibliography Type:	Description: (Last Updated: 01/03/2026)
Abstracts for Journals and Proceedings	Stoltz A, Ratri A, Hong X, Christenson LK. "Effects of space flight and microgravity on adrenal tissue." 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020
Abstracts for Journals and Proceedings	Stoltz A, Ratri A, Hong X, Christenson LK. "Effects of Space Flight and Microgravity on Adrenal Tissue." K-INBRE Regional Symposium, Virtual, January 14-15, 2021. Abstracts. K-INBRE Regional Symposium, Virtual, January 14-15, 2021. , Jan-2021

Articles in Peer-reviewed Journals	Hong X, Ratri A, Choi SY, Tash JS, Ronca AE, Alwood JS, Christenson LK. "Effects of spaceflight aboard the International Space Station on mouse estrous cycle and ovarian gene expression." npj Microgravity. 2021 Mar 12;7:11. https://doi.org/10.1038/s41526-021-00139-7 ; PMID: 33712627; PMCID: PMC7954810 [Note reported previously in Jan 2021 as "in review as of January 2021"] , Mar-2021